1. Multi-Agent Reinforcement Learning in a Competitive Game — Project Brief

Objective:
Implement and analyze multi-agent RL algorithms (Independent Q-Learning, MADDPG, or QMIX) in a competitive environment like Predator–Prey or Pong Doubles.

Motivation:
MARL introduces non-stationarity and coordination challenges that do not appear in single-agent RL. Students gain insight into emergent behavior and stability issues.

Core Tasks:

- Build or adopt an existing multi-agent game environment.
- Implement at least one cooperative and one competitive MARL algorithm.
- Train agents under varying difficulty levels.
- Evaluate convergence, stability, and agent strategies.

Expected Outcomes:

- Analysis of credit assignment and non-stationarity.
- Comparison of MARL algorithms.
- Insights into emergent behavior.

Deliverables:

- Code repository
- Report in IEEE conference format (4–6 pages)
- Presentation in the format attached to the mail

2. POMDP & Belief-State Learning with DRQN — Project Brief

Objective:
Train agents in a partially observable environment using recurrent RL (DRQN or LSTM-DQN), and compare performance to standard DQN.

Motivation:
Many real-world problems lack full observability. Students learn how memory helps maintain useful belief states.

Core Tasks:

- Design a partially observable grid world (e.g., occlusion, noisy sensors).
- Implement DQN baseline & DRQN with LSTM.
- Analyze how internal memory affects performance.

Expected Outcomes:

- Quantitative comparison of DQN vs DRQN.
- Visualization of hidden-state dynamics or belief formation.

Deliverables:

- Code repository
- Report in IEEE conference format (4–6 pages)
- Presentation in the format attached to the mail

3. Hierarchical Reinforcement Learning (Options / HIRO) — Project Brief

Objective:
Apply HRL to long-horizon tasks like Ant Maze or Taxi domain, using Options Framework or HIRO.

Motivation:
HRL enables temporal abstraction and improves long-horizon credit assignment.

Core Tasks:

- Implement high-level and low-level policies.
- Define or discover subgoals.
- Compare HRL performance against flat RL.

Expected Outcomes:

- Increased sample efficiency.
- Evidence of meaningful sub-policies.
- Analysis of subgoal usefulness.

Deliverables:

- Code repository
- Report in IEEE conference format (4–6 pages)
- Presentation in the format attached to the mail

4. Offline RL on the D4RL Benchmark — Project Brief

Objective:
Use offline RL algorithms like CQL or IQL on D4RL datasets (HalfCheetah, Hopper, Walker2d).

Motivation:
Offline RL is essential for safety-critical domains and real-world applications where data collection is expensive.

Core Tasks:

- Load and preprocess D4RL dataset.
- Train CQL/IQL agents.
- Compare with behavior cloning baseline.
- Evaluate OOD generalization.

Expected Outcomes:

- Evaluation on normalized D4RL scores.
- Insights into distribution shift & conservatism.

Deliverables:

- Code repository
- Report in IEEE conference format (4–6 pages)
- Presentation in the format attached to the mail

5. Safe Reinforcement Learning with Constraints — Project Brief

Objective:
Train an RL agent to navigate a hazardous gridworld while satisfying safety constraints using Lagrangian RL, Safety Layer, or Shielding.

Motivation:
Safety is critical in autonomous systems like drones, robots, and smart grids.

Core Tasks:

- Build safety-constrained environment.
- Implement at least two safe RL methods.
- Compare reward vs. violations trade-offs.

Expected Outcomes:

- Violation rate plots.
- Pareto analysis: performance vs safety.
- Explanation of which method handles constraints best.

Deliverables:

- Code repository
- Report in IEEE conference format (4–6 pages)
- Presentation in the format attached to the mail

6. RL for Energy Management in Smart Grids — Project Brief

Objective:
Use RL to minimize energy cost and peak loads in a simulated microgrid with batteries, renewable sources, and variable demand.

Motivation:
Smart grid control is a real-world use case with stochastic dynamics and multi-objective constraints.

Core Tasks:

- Build/load microgrid environment (battery, load profile, solar/wind inputs).
- Train an RL agent using DQN, PPO, or SAC.
- Incorporate multi-objective reward shaping.
- Evaluate energy savings vs operational constraints.

Expected Outcomes:

- Demonstration of cost savings.
- Trade-off curves (energy cost vs constraint violations).
- Policy interpretation under uncertainty.

Deliverables:

- Code repository
- Report in IEEE conference format (4–6 pages)
- Presentation in the format attached to the mail

7. Model-Based RL with Learned Dynamics — Project Brief

Objective:
Implement MBPO or PETS to solve continuous control tasks (CartPole, Reacher, Pendulum swing-up).

Motivation:
Model-based RL improves sample efficiency through learned dynamics and planning.

Core Tasks:

- Train neural network dynamics model.
- Use MPC or rollout-based planning.
- Compare model-based vs model-free RL.

Expected Outcomes:

- Faster convergence curves.
- Visualization of model error over time.
- Study of performance degradation due to model bias.

Deliverables:

- Code repository
- Report in IEEE conference format (4–6 pages)
- Presentation in the format attached to the mail

8. Curriculum Learning for Hard RL Environments — Project Brief

Objective:
Design a curriculum that gradually increases task difficulty, and study its impact on agent learning.

Motivation:
Curriculum learning stabilizes training for sparse-reward or complex tasks.

Core Tasks:

- Create a progression of tasks (easy → moderate → hard).
- Train agents with and without curriculum.
- Compare convergence, sample efficiency, and stability.

Expected Outcomes:

- Learning curves demonstrating curriculum benefits.
- Insights into curriculum scheduling.
- Discussion on transfer between stages.

Deliverables:

- Code repository
- Report in IEEE conference format (4–6 pages)
- Presentation in the format attached to the mail

9. Meta-RL for Fast Adaptation (MAML / RL^2) — Project Brief

Objective:
Implement a meta-RL method that enables rapid adaptation to new tasks.

Motivation:
Meta-RL teaches the agent *how to learn*, not just *how to act*.

Core Tasks:

- Create a distribution of related tasks (mazes, bandits, navigation goals).
- Implement MAML or RL^2.
- Test on unseen tasks.

Expected Outcomes:

- Significant reduction in adaptation episodes.
- Analysis of the internal "learning dynamics."
- Meta-train/meta-test performance comparison.

Deliverables:

- Code repository
- Report in IEEE conference format (4–6 pages)
- Presentation in the format attached to the mail

10. Explainable Reinforcement Learning — Project Brief

Objective:
Analyze RL agent decisions using interpretability tools like saliency maps, feature attribution, policy distillation, or decision-tree extraction.

Motivation:
Real-world RL deployment demands transparency, especially in safety-critical systems.

Core Tasks:

- Train an RL agent (Atari / Gridworld).
- Apply at least two interpretability techniques.
- Visualize and discuss learned behavior.

Expected Outcomes:

- Heatmaps showing attention regions.
- Simple interpretable policy model (if distilled).
- Explanations of failure vs success cases.

Deliverables:

- Code repository
- Report in IEEE conference format (4–6 pages)
- Presentation in the format attached to the mail